

Expert Recommendation No. 3 of 16 June 2020

Safety and tactics in vegetation fire operations¹

1. General information

Fighting vegetation fires usually involves a commitment to preserving the utility, protection and recreational functions of forests, Bush or heathland or agricultural land.

Above all, the forest serves as a drinking water reservoir, oxygen supplier, CO₂ consumer, as a timber supplier, extreme reducer (heavy rain (flood protection!), heat, storm, noise and other) and last but not least as a (local) recreation area for industry company.

Even in the case of forest and other vegetation areas, a distinction is made between the A distinction is made between preventive and defensive fire protection. Preventive forest fire protection, which primarily falls under the responsibility of the state forestry ministries or the respective forest owners, focuses on preventive measures. These include, among other things, the creation and maintenance of protective strips, forest fire barriers, fire water extraction points, and year-round accessible paths, early (forest) fire detection, and, last but not least, targeted information campaigns in the media, for example, on the current forest or grassland fire risk.

Defensive forest fire protection, which falls under the responsibility of the state interior ministries or fire protection authorities, is characterized by the classic ground and air-based firefighting methods out of.

When fighting vegetation fires, the safety of the to give emergency services the highest priority in operational planning,

¹ This technical recommendation is based on the versions first published in 2007 and updated in 2018. It has been revised in terms of content, taking into account operational experience from the forest fire years and the increasing use of aircraft.

This means that in all measures the risk to the emergency services must be as low as possible as possible.

Principle of vegetation fire fighting

The goal must be to get a fire under control as quickly as possible.

A safe and effective use of all forces and resources is essential to note!

In order to be able to orientate oneself better in the event of extensive vegetation fires and in particular to be able to communicate clearly with deployed aircraft standards must be defined.

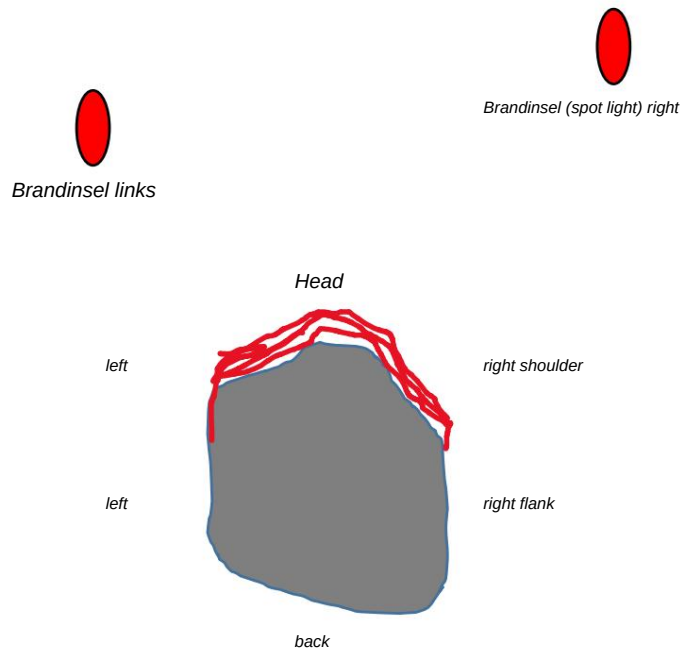
These begin with the coordination of a uniform map or Coordinate system (fire departments generally use UTM coordinates, while aircraft mostly use GPS data based on the WGS-84 system). If a uniform system is not possible, the "map datum" used for the location coordinates must be transmitted so that each unit can convert them if necessary.

The north position and map classification must be clearly determined and always kept available in order to be able to react clearly to changes in the situation (for example, wind changes direction and strength).

Directions are always to be given in degrees (clockwise).
ben.

If no compass is available, directions can be obtained from a
an axis of two reference points (for example, a prominent church tower – which is clearly recognizable tower of a wind turbine) in the estimated position of a covered
For example, "from the direction of view of the
Church tower to the wind turbine, from there at two o'clock, smoke development."

The areas of a vegetation fire are designated as follows:



In the case of vegetation fires, a distinction is also made between

- Fire in the area of the ground (deep earth or humus, superficial ground, bush and trunk fires) and
- Fire in the trees (full fire, crown fire)

differentiated.

The type of fire and its further development depend on the vegetation (stand conditions, stocking density, tree species, age classes), the topographical terrain conditions and the meteorological conditions (temperature, wind direction and strength as well as the precipitation balance and thus also the air humidity).



1.1 Influence of vegetation

When considering the influence of vegetation on fire protection, some important features can be listed:

- Conifers (pine, spruce, fir, larch, etc.) are particularly vulnerable to fire. than deciduous trees (beech, oak, maple, chestnut, ash, etc.). A special The birch is a deciduous tree, but its bark and even green leaves become inflamed much more quickly in extreme dryness, like those of other deciduous trees.
- Deciduous trees are used in today's forestry in areas particularly prone to forest fires. endangered areas as natural fire-retardant barriers. plants.
- Especially young conifers (thickets, poles) are particularly susceptible to fire and especially to full-blown fires due to their branches that reach deep into the ground.

- Due to their low water retention capacity, sandy soils are generally less likely to pose a vegetation fire risk than water-rich clay soils.
it.
- With the beginning of the spring months, the risk of forest fires (number of forest fires) is particularly high due to the “dried-out vegetation”.
- Fires of dry reeds spreading very quickly with the wind
and heathland areas occur mainly in spring.
- Fires on dry, cultivated land (cereals, oats, rye, etc.) naturally occur more frequently at the end of summer or in
Autumn ahead.
- In the continental areas of East Germany (hot and low-
low-fire summer months), the vegetation fire risk is related to the
Number and total area of damage usually higher than in the other parts
Germany. Nevertheless, experience shows that in the case of prolonged dryness
in almost all forest-rich federal states in the last decades.
large vegetation fires were also recorded.

In the context of operational tactics, it should always be taken into account that the
Fire spread to areas with an increased risk of forest fires
In case of emergency, these
It is preferable to seal off endangered areas, as an effective fire-fighting attack
only possible with a large expenditure of personnel, technology and extinguishing agents
can be.

1.2 Influence of the terrain and the subsoil (soil)

Even though the areas with high forest fire risk are located in the German
Lowland plains (for example Lüneburg Heath, Brandenburg etc.) are located,
Fires in hilly or mountainous areas can pose a particular danger.
place.

Fires spread in mountainous terrain due to thermal radiation, ther-
mik and possibly the updraft upslope particularly quickly, therefore
is a fire-fighting attack or the laying of containment lines above fire edges

extremely critical. The rate of fire spread is often significantly reduced.
Added to this is the risk of falling and the danger of loose stones or tree trunks in mountainous areas!

Fires in dried-out or former moorland (heath) can be
The fire can spread very quickly on a mostly flat surface, driven by the wind.
develop into the combustible soil layers. As a result, it can
to the falling of trees without any discernible warning or the collapse (sagging) of
vehicles and people into burnt-through soil layers.
men!

In order not to endanger the safety and health of the emergency services, the influence of the terrain and the subsoil must be taken into account when planning and carrying out the emergency measures (see also Section 15 (1) of DGUV Regulation 49).

1.3 Influence of meteorology

In addition to the biotopic conditions of a forest stand (tree species, age, stocking, ground cover, etc.) and the geographical location, the meteorological parameters

- Wind speed and direction,
- Humidity or dew point temperature and
- Precipitation (precipitation totals, precipitation forecasts)

essential for the initiation, spread, and suppression of vegetation fires. A meteorologist should be involved in the operational management of large forest fires, especially to determine the weather situation and predict weather developments (meteorological conditions).

Statements on forest fire risk forecasts can be found, for example, at

<http://www.wettergefahren.de/warnungen/indizes/waldbrand.html>

Always pay attention to the grassland fire index, because this is especially important in Moor and heath areas as well as for fires on agricultural land and in the reeds is much more meaningful:

<https://www.dwd.de/DE/leistungen/graslandfi/graslandfi.html>

It should be noted that such vegetation fire danger indices only serve as a rough estimate of the general meteorological risk potential. The actual forestry risk on site may vary for various reasons. This is due to various factors, such as storm and drought damage, deadwood due to insect infestation and logging debris, as well as local, primarily topographically influenced weather conditions.

The indices represent a mathematical combination of different weather parameters, which, however, as individual factors in the case of use, should not be underestimated. of importance (for example, wind speed and wind direction).

1.4 Tactical priorities

In contrast to fighting building fires, the first priority is fighting vegetation fires as quickly as possible of the fire source. A complete and immediate extinguishing of the entire area is only possible in the rarest of cases and initially not tactically feasible by the upper ter priority.

The containment of the fire should only be carried out in the case of controllable fires with the begin to fight the fire front and secure the flanks, if this is the intensity of the fire without endangering yourself.

In the case of larger and dynamic vegetation fires, proceed as follows:

- 1) Define and secure one or more anchor points (for example, one on each flank).
- 2) From there, carry out the fire-fighting attack over both flanks.
- 3) Comprehensive firefighting attack on the flank, containment of the front via the shoulder and

4) Comprehensive extinguishing work along the entire perimeter.

If this is not possible, the player must be in a tactically advantageous position (favorable in terms of the vegetation and terrain as a natural holding or catch line) attempts to stop the fire front.

The protection of objects (buildings, industrial facilities, important infrastructure lines such as electricity and gas lines) usually takes priority over the protection of vegetation areas. Of course, the fire development, the own Possibilities of combat in terms of tactics, but also in terms of a safe your retreat.

In buildings, all windows and doors must be closed before a fire front arrives. Close, special fire loads around the buildings (for example: equipment, vehicles, etc.) must be removed as quickly and thoroughly as possible. If this is not possible, attempts can be made to achieve a certain degree of protection against ignition by permanently wetting the area with water (for example, by means of stationary irrigation systems) or by applying a layer of foam or gel.

Full fires, especially when the crowns have ignited and the fire wind-driven, can practically only be fought defensively!

Breaking an approaching full fire front is only possible with massive water use possible from the side or from above.

2.1 Use of fire engines

Especially in the early phase of forest and wildfires, the available and scarce resource of fire-fighting water was used extremely efficiently. If the extinguishing agent is water with an unnecessarily high application rate is output, the extinguishing water supply is used up too quickly. The use of water cannons in particular must therefore be examined critically.

For extinguishing ground fires, which account for the majority of forest fires in Germany, the use of C and D pipes can certainly be (basically a spray jet). This is especially true for reasons of "Maneuverability" of the lines in the terrain and the hoses contained required amounts of water (C 15/42 about 21 liters).

Water from jet pipes should only be used to treat fire compartments with open flames and An intensive flame development must be combated. A complete removal and Extinguishing the fire edge and especially the entire affected fire area will only be treated in a later step with another fire engine or Firefighting teams may be possible.

Driving over fire lines can damage the vehicle's control or supply lines (e.g., compressed air, fuel, or hydraulics), causing the vehicle to stop on the fire line. The vehicle must therefore be specially designed for this type of firefighting. To date, only a few such vehicles exist in Germany (for example, in Brandenburg or with the German Federal Armed Forces Fire Service). Until certainty is established, it should be assumed that the (firefighting) vehicles are unprotected against flames! The same applies to protecting the vehicle engine's air filter from ignition by flying sparks.

If you look at a fire from behind with the wind, that is from the burnt side, the actual fire-fighting operation must first be carried out by setting up place of the vehicle.

If there is no safe turning possibility in a forest path or building access journey is available or can be created directly during the operation, must be retracted backwards in order to be able to retreat quickly if necessary (own security).

For your own safety, it should always be possible to use an additional C or D pipe.
or the S-pipe, therefore not all pressure outlets should
and always about 300 to 400 liters of water are retained in the tank.

If the fire is attacked from the front, the vehicle should be positioned in the direction of escape.
Due to the inability to detach the hose couplings from the vehicle in case
of a retreat, the firmly screwed, dimensionally stable rapid attack
hoses pose a risk. Roll-up hoses can be easily uncoupled,
dividers can be used over a wider area and have longer
Hose sections have a much lower pressure loss.

Fire edges with intense flame formation (flames with a length of more than about two
meters) should not be attacked head-on, but rather from behind or from the sides. This
minimizes the risk of
Part of the fire's edge is likely to become trapped. If the area is confined and there is a
risk of a full-scale fire, a targeted, quick, and short-term application of more water, even
from pipes with a greater throw range (e.g., from a water sprinkler), may be appropriate.

Tank fire engines specifically designed for ground-based forest firefighting must be
able to deliver extinguishing water in a controlled manner while driving ("pump and roll")
and be designed for off-road use. Furthermore, they must, at least in the future, be
equipped with appropriate mechanical and thermal protection for the lines and protection
of the air filter against ignition by
be equipped with flying sparks.

Small, maneuverable tank fire engines or specially developed forest fire
Tank fire engines should be used as attack vehicles, large and heavy tank fire engines
vehicles are used as feeder vehicles. A water transfer point with
appropriate buffer capacity must be determined.

The use of wetting agents increases the effectiveness of the extinguishing water used and thus meets the generally prevailing lack of this and is therefore a must.

Air foam, compressed air foam and gelling agents are particularly suitable for To secure clearings, wound strips and objects.

Never leave vehicles with hot exhaust systems on dry fields or similar areas! There is a risk that the dry vegetation ignited by it.

This applies not only to all vehicles with catalytic converters located on the vehicle floor (many cars with petrol engines), but also to an increasing number of Euro VI vehicles in the future, which, in addition to often very hot exhaust systems, can either be manually switched to regeneration mode depending on the chassis programming or, if there is no special vehicle programming, may be automatically switched to this mode. switch.

After use, check whether there are any embers or smoldering residues on the chassis or in the area of the engine (e.g. air filter). could have collected!

Never block access to forest roads with emergency or assigned vehicles, as these can serve as attack and, if necessary, escape routes for units. These escape routes must be marked and

The same applies to security zones.

Suitable storage or assembly areas must be explored, assign and use.

2.2 Deployment of firefighting teams on the ground

Through the deployment of trained firefighting teams, which, among other things, Equipped with backpack sprayers/fire extinguisher bags, fire beaters and shovels

are, the wind can create fire edges with flame lengths of up to one to two meters high.

Equipment and tactics differ between the operational scenarios
Forest fire and field fire.

In the forest, fires can be extinguished using backpack sprayers/fire extinguishing bags and sand throwing.
sections with particularly intense flame formation on the ground.
Remaining fire and less intensely burning fire edges can be extinguished with
Fire beaters and other suitable hand tools can be used to combat the fire.

In fields, due to the strong root penetration of the soil, obtaining sand is usually too time-consuming; the use of backpack sprayers and fire beaters is most effective here.

The following safety instructions must be observed, particularly when emergency personnel are deployed at a fire.

2.3 Personal protective equipment

When working near fire edges, protective clothing must offer a good compromise between protection against brief flame contact, flying sparks and the best possible wearing comfort.

The use of multi-layer fire protection jackets and trousers according to DIN EN 469 is too much of a burden for the wearer in warm weather.

The use of a single-layer fire protection jacket and trousers is recommended for example, the PSA 11 according to DGUV Information ~~205-014~~ (see also §14 DGUV Regulation 49).

For work on the fire edge, the firefighting teams should also be accompanied by a
Flame protection hood.

A fire helmet (as light as possible!) with neck scarf, suitable protective gloves
Shoes and boots are essential PPE.

Wearing respiratory protection (respiratory protection connection with filter or air-independent breathing apparatus) is usually not required or burdensome.

puts too much strain on the wearer. A particle-filtering half mask (e.g. FFP2 according to DIN EN 149) or forest fire scarf with integrated filter in conjunction

With tight-fitting safety goggles, working in slightly smoky areas.

However, as an escape device, a breathing mask with a filter (CO- or Combination or "firefighter filter" A2B2E2K2-P3, but better with A2B2E2K2-P3-CO) should be carried. For longer and heavy physical work
However, these filters are not suitable.

The intentional wetting of protective clothing is prohibited due to a possible
Hot steam penetration is dangerous and should therefore be avoided!

2.4 Protection against heat exhaustion

The duty of fire service leaders to provide care and maintain the performance of the deployed emergency personnel in accordance with Section 3 (3) of DGUV Regulation 49 must be observed, for example by timely replacement, sufficient breaks, effective hygiene, and suitable catering (food and drinks).

Since ground-based forest firefighting often takes place under difficult conditions, conditions where heavy physical work has to be carried out, it should be and drink enough after use. The rule of thumb here is about one liter per hour.

Physiologically, it makes more sense to take a few sips every ten minutes, instead of gulping down a liter after an hour – the body can

Drinks should be non-carbonated water or
Fruit juice spritzer can be used.

Each emergency worker should have a personal supply of drinking water, for example in
Carry canteens or drinking systems with you.

2.5 Working and extinguishing equipment, units, notes

The composition of the work and extinguishing equipment for firefighting teams depends
heavily on the existing vegetative site conditions.

hear among other things

- Hoes (for example, so-called fire rakes or hoopoe hoes)
- Shovels ("Bavarian sand shovels") are used for digging out soil and
suitable for targeted throwing and available everywhere, of course other
similarly shaped shovels or "forest fire shovels" can be used
it).

Not so suitable are pure sand shovels with a semi-circular or straight front
Leaf, because with this you can get a bit firmer soil with some grass roots
can no longer pierce well.)

- Forest fire or fire beaters
- Chain saws with suitable PPE and accessories
- Backpack sprayers or fire extinguishing backpacks or similar
- D-hose material, C-DCD distributors, simple D-hollow jet nozzles
- Carrying frames for equipment and supplies

The additional load kit "J" for forest fires, which has been standardized for several
years according to DIN 14800-18, provides a good basic equipment for this
purpose. It can and should be supplemented regionally according to existing
special requirements, for example, with special hoes or axes or more hose material with portable
small motor pumps.

In many places, special units are set up to fight forest fires depending on the risk. provided and maintained, such as fire protection units, forest fire combat platoon (also with mobile liaison units [motorcycles]), fire water supply train (with, for example, KdoW/ELW 1, 2 x LF 20-KatS, SW 2000), fire water transport train (with, for example, KdoW, GTLF, TLF 4000, WLF with AB tank) etc.

Equipment specifically for forest fire fighting depending on the individual risk must be provided for. Alternative means of deployment such as forestry and clearing vehicles, construction and agricultural machinery (wheel loaders, ploughs, Milling machines, slurry tankers!) must be included in the operational preparation.

A logistics unit with the possibility of on-site fuel supply and at least field repairs (for example, tire changes) should be from association strength to each unit.

Prepared forest fire protection maps with information on

- Forest and forestry roads accessible to emergency vehicles
- exact place names
- Forestry, rescue and assembly points
- Possibilities for extracting fire water
- Areas suspected of containing explosive ordnance
- Areas at risk of full fire (e.g. with coniferous trees, damage/disaster community areas)

are essential and in sufficient numbers (if possible also in digital form).

2.6 Use of aircraft

In principle, it can be stated that aerial vegetation fire fighting always serves only as a supporting measure to ground-based forest fire fighting.

Firefighting water drops from aircraft can be used, among other things, to quickly
However, this must be done with the use of fire engines
and teams must be coordinated so that a suppressed fire
cannot develop again.

Before dropping, ensure that no persons (emergency personnel/non-emergency services) are in the drop area.

When approaching straight and dropping in a line, a range of 20 to 30 m lateral
The safety distance depends on the flight axis.

- Flight speed (here especially rotary-wing aircraft = helicopter or fixed-wing aircraft
= fixed-wing aircraft)
- Flight altitude
- Container type and opening as well as
- Firefighting area (stand type, age of the stand, trees/branches that
can break through the drop)

Emergency vehicles can also provide a safe place to retreat, but may be damaged
by the drop.

The discharge must be carried out in such a way that with each throw, a portion of
the quantity (as seen across the width of the throw) is directed into the burning area
and a portion onto the not-yet-directly-burning area along the fire line. This provides
sufficient safety in the never-completely straight line of fire, cools already heated
range and extinguishes any individual flames that may be ahead of the line.

In the case of longer fire fronts or flanks, the drops must be
The control of multiple aircraft must be coordinated accordingly.
tactically sensible.

Therefore, at least temporarily, a tactical division of the input
sections by a different selection of resources per section

be possible, for example left flank: helicopter, right flank: ground units
ten).

Nevertheless, it must be ensured that immediately after setting the
Airborne drops restore ground-based forest fire fighting
be continued immediately, otherwise effective deletion success cannot be guaranteed.
that can.

This also requires secure communication!

It is always advisable to use a tactical coordinator who is
Ground in direct communication with the aircraft coordinates the exact drop target,
evacuation and resumption of ground-based firefighting operations.
dined.

For larger or more complex situations, an Aircraft Operations Section Leader (EAL
Air) should also be deployed. This leader advises the operations commander on the
aircraft deployment options and then implements the defined measures (e.g., aerial
firefighting with three helicopters on the right flank). Their responsibilities include:

- The selection of landing sites in consultation with the operations manager and the needs of aircraft.
- The establishment of the landing sites including the necessary special logistics for each as required (e.g. supply, refueling, fire-fighting water, organization tion of material and personnel transport).
- Securing flight operations as required (e.g. crash team at the landing site, water rescue in open waters, firefighting water intake...)

When water is dropped by aircraft, the force and weight of the water can seriously injure emergency personnel. Therefore, the additional use of aircraft in forest firefighting operations must be closely coordinated within the technical operations command.

A special case of firefighting from the air (and in these cases often the only possibility of a direct attack) is to fight fires on Areas suspected of containing explosive ordnance (military training areas or areas). However, in addition to the fire service, Service Regulation 500 also includes the distance rules defined by the explosive ordnance disposal services (DGUV Rule 113-016 "Blasting work")!

Fighting forest fires in very difficult terrain, for example on steep mountain slopes with rotary-wing aircraft is a fundamentally possible option, However, it is very demanding. This work involves additional risks due to from poorly visible obstacles (e.g. cable cars) and weather conditions conditions (downdrafts, thermals, funnel effects). Therefore, even more experienced pilots with special mountain training are required.

When using external fire water tanks on helicopters, please note

Please note that not all containers (foldable/rigid external cargo containers) can be flown by all helicopter types (weight limits/towing capacity depend on the individual helicopter type and its specific equipment or fuel supply), and not every helicopter type is readily available. Furthermore, the options for transporting loads can change significantly depending on the required flight altitude. The higher the location and the more fuel the helicopter still carries, the smaller the loads that can be flown.

The use of rotary-wing aircraft with external load containers (ALB) or other aircraft (e.g. for observation and command support) therefore requires intensive supra-regional preliminary planning. This requires clarifying:

- Who can carry out such an operation (federal, state police, federal the armed forces or other armed forces, possibly private helicopter operators before)?
- Which ALB in which sizes are available and can be used with the Available helicopters and the existing suspension systems can also be "flown" (rigid external load containers (e.g. LAB 5000)/Au- External load folding container (for example Bambi Bucket)?

- Which fixed or foldable containers on the ground are used, if any?
requires
 - o Water extraction from this container by helicopter (in addition to larger folding containers, waterproof roll-off containers are also suitable, for example AB Transport, collection or similar?
 - o Water delivery by helicopter into this container (here are especially sturdy, foldable containers)?
- What regional deployment time/readiness is achievable after the preliminary planning?
cunning?
- Are regional support units necessary or required for this?
seen?
- How should the alarm be raised in the best possible and timely manner; operations are usually subject to a charge.
- How can the coordination of aircraft deployment be achieved? (This requires
The technical means of communication are also available and an integration
transfer to the tactical operational command should be possible.)

When using aircraft, the use of extinguishing agent additives should also be (especially wetting agents) should be provided.

Likewise, external fire water tanks can be used to shuttle to open portable (foldable) fire water tanks in remote areas with poor water supply. Should lifting screwdrivers are used in shuttle traffic, so transportable, depending Depending on the external load containers used, sufficiently large and stable open folding container required.

The parallel use of

- drones and manned aircraft or
- Fixed-wing and rotary-wing aircraft

in an operational area is too dangerous and should therefore be avoided. Exceptions apply only to aircraft with observation or Coordination tasks at a significantly different level.

The discharge and transport of fire water from aircraft must always be well (pre-) If he misses the target, he can not only cause injuries/damages For example, damage to emergency personnel or vehicles; the requirement of a efficient use of firefighting water.

In otherwise difficult or inaccessible areas, it is possible to use helicopters ber also carry out the transport of external loads and personnel. For this purpose, the relevant guidelines for the transport of loads must be observed. become.

2.7 Follow-up extinguishing work and fire inspection

Vegetation fires, especially ground fires, often involve deeper, flammable, not yet completely decomposed soil layers. Especially in former moorland areas, the fire can even spread underground and reemerge elsewhere. Embers can also be found in root fragments.

The next stronger wind can then fan the fire again. This Because the stock has already been attacked by the fire, it can initially spread even faster than the original, supposedly extinguished, fire.

This means that the irreplaceable fire-fighting work and the fire inspection (one (general fire watch) are complex and time-consuming.

The main tactical focus is therefore initially on securing the entire burn area edge, i.e. the area between burnt ("black") and unburned ("green"). The securing work should begin here at the front (i.e. with the wind direction), because here the greatest propagation relation- There is a risk of re-ignition.

Depending on the situation on site, available fuel on both sides of the fire line never, soil type, depth of the fuel (peat), weather (wind) and size of the entire fire area (clarity), the area to be secured should be measured width and depth. A recurrence must be prevented.

This can also be done with machine support, for example by Forest fire or planting ploughs, tillers, wheel loaders or mulchers be used.

Afterwards, if possible, you should also remove the embers in the burnt area search, open and extinguish or cover by working into soil or sand and suffocate.

For larger fire areas, it is hardly possible or sensible to dig up every square meter with a C-hose or hand tools. Experience has shown that there are certain areas where firefighters need to extinguish fires. Rootstocks/root areas (shrubs, grasses [reed grass]), areas with fallen wood (trunks), ditches, and generally thicker litter layers (e.g., coniferous litter), logging debris (e.g., cut branches, treetops), and not completely rotted humus layers (e.g., peat) are considered ember-holding points and should be addressed first. When securing the fire area, hose lines (C or D) can be used in full jet mode if sufficient water is available or hoses can be laid there. Particularly in inaccessible areas, it is advisable for teams to use a combination of hand tools (e.g., hoop hoe, spade) and firefighting backpack or backpack.

syringe to use.

If the fire area is to be secured in the evening, night and morning hours, suitable and sufficient illumination of the area of operation is required. beets.

Regular inspections within the fire watch are advisable. This must be long time until all embers are safely found and extinguished.

For this purpose, it is advisable to use high-resolution thermal imaging cameras, among others, from aircraft (for example, police) or drones.

3. Safety rules

For every use in larger forest and area fires, we recommend, in addition to the requirements of DGUV Regulation 49, the following generally applicable safety regulations:

- Proceed only in units or clear sub-units (squad, staff-field, group, train).
- Maximum distance of the emergency services within sight of the next emergency services
- At least one portable radio per deployed unit
- Signal whistle or radio for each emergency worker
- Check access to the site before driving (e.g. height, Width, condition, turning ability, axle loads) and hazard situation in Be-
Check the fire development.
- Contact with neighboring emergency services or units in the section or even search beyond the section boundaries, in order to be able to detect changes in the situation at an early stage.

There are situations where increased caution is required:

- Attacking a fire edge in a confusing situation
- Attacking a fire edge without sufficient reconnaissance
- Frontal attack on a fast-moving fire edge
- No connection to the operations management
- Wind suddenly picks up or changes direction (for example during thunderstorms).
tern)
- Fires caused by flying sparks and flying sparks behind your own position
tion
- Fire below your own position
- Beginning of darkness
- Water drops by fixed-wing/rotor-wing aircraft

- Danger of landslides when fighting vegetation fires in recultivated fourth mining areas
- Suspicion of explosive devices, at the latest when there are unclear flame events nuisances (for example by burning phosphorus ammunition), explosions or that gives something similar.

In addition, the following aspects must be considered. Internationally, The acronym **LACES** has been adopted, which also translates well into German can be transferred.

3.1 L for Lookout = observation post

An independent observer should be present at each operational section to sudden weather changes (wind) or fire islands (spot fires) the To do so, he should use suitable equipment (binoculars, weather sensing devices, whistle or horn). In order to change even under difficult conditions a quick evacuation of the

In order to be able to carry out a rescue operation in the endangered area, a clear retreat signal should be (for example, pressing the vehicle horn three times briefly, pause, repeat tion, etc.) can be agreed upon.

3.2 A for Anchorpoint = secure anchor points

Every safe operation begins with one or more anchor points. These can be non-flammable areas (gravel surfaces, wide roads, crossing Land/water) or safely extinguished areas (a sufficiently large, definitely area protected from flashback).

3.3 C for Communications = Ensuring communication

A connection to the operational command and within the unit is required for forest and If it is not there, there is a great danger of sudden to be surprised by changes in the situation.

Therefore, the aircraft used must also be securely integrated into the communication be involved at the site.

Appropriate communication plans according to FwDV 800 must be drawn up and to keep the entire operational process up to date.

3.4 E for Escape = ensure escape/rescue route

Each operational section must have a previously selected escape route.

This escape route must be known to all emergency services and should, if possible, have a low fire load and lead to a safety zone.

If it is not clearly identifiable, it must be marked.

Marking procedures must be known to all
be!

3.5 S for Safety = explore or create a safety zone

A safety zone is an area in which the emergency services can operate without further Protective measures against a fire are safe and a fire edge prevents the emergency personnel. This can be a road, a rocky area, or a body of water. If necessary, this must be created by removing vegetation.
it.

The necessary size of the safety zone is determined by the total number of emergency and support personnel deployed in this operational area or section!

4. Handover of the site

After the firefighting has been completed, the site is handed over to the owner. mer or his representative(s) with the aim of transferring the tasks of a Fire watch must be conducted. This must then be documented in the incident report.

Notes

This technical recommendation will be supplemented in the coming years by additions to the Partial areas of the operation were expanded and extended, in particular to to better coordinate cooperation with the other parties involved.

Fire brigades are advised to use their equipment within the framework of their conditions (geography, vegetation, weather conditions, equipment) and to determine the operational related training based on this specialist recommendation.

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This expert recommendation was made by the Forest Fire Working Group in the German Fire Brigade Association in close coordination with the Technical Committee on Operations, Extinguishing resources and environmental protection.

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Disclaimer: The expert recommendation "Safety and tactics in vegetation fire operation" was carried out to the best of our knowledge and with the greatest care by our Experts and approved by the relevant departments and the DFV Executive Board
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